

MODEL QUESTION PAPER

TED (10) 4094

SIXTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

PLANT UTILITIES AND MAINTENANCE

Maximum Marks:100

Time:3hrs

PART- A

I Answer the following question in one or two sentences

(Each question carries 2marks)

- 1 what is meant by dry bulb temperature
- 2 Define dryness fraction
- 3 Define saturation temperature
- 4 What is meant by hydro blasting?
- 5 Define ventilation

2X5=10

PART -B

II Answer any five of the following (Each question carries 6marks)

- 1 What are the factors to be considered for the selection of cooling towers?
- 2 Explain the preparation of boiler feed water.
- 3 Explain the working of Benson boiler
- 4 Draw the neat sketch of blow off cock valve and mark their parts
- 5 State the scheduled maintenance and its advantages
- 6 Explain the mechanical cleaning method of tubular equipments
- 7 Draw the typical master plot plan of a chemical plant ,you know

6X5=30

PART –C

Answer one full question from each unit (Each question carries 15 marks)

UNIT I

III a The dry bulb temperature and dew point of ambient air were 5X2=10

found to be 35 ° C and 20 ° C respectively. The barometer reads 101-325 kN/m². Calculate (1) Molal humidity (2) Absolute humidity (3) Relative humidity (4) Saturation humidity (5) Humid volume and Humid heat. Data given vapour pressure at 35 ° C 99.9 kN/m². VP at 20 ° C 4.3 kN/m²

- b State fire fighting water. 5

OR

- a Explain the constructional details and working of dehumidifier 7

- b A Weather bureau reports dry bulb temperature 39 ° C and relative humidity of 50% . The barometer reads 740 mmHg Find the percentage humidity of air. Vapour pressure of water at 39 ° C is 50 mm Hg 8

UNIT II

- a Explain the constructional details and working of vertical boiler 9

- b Explain the working of pressure gauge. 6

OR

- a Explain the constructional details and working of Cochran boiler. 10

- b Explain the working air pre-heater 5

UNIT III

- a Explain the common trouble and remedial action of filter press 10

- b State the cleaning methods of towers 5

OR

- a Explain how preventive maintenance is better than break down maintenance. 10

- b State the objective of maintenance 5

UNIT IV

- a What about your opinion of site selection of Cochin refinery? Justify your answer 12

- b Enumerate the commonly used roofing materials in chemical plant 3

OR

- a Explain the product lay out and enumerate its merits and demerits 10

- b Describe the safety factors involved in plant lay out. 5

